

SVKM's Narsee Monjee College of Commerce and Economics

(Empowered Autonomous)

Internet of Things

A project submitted to the Department of BSc.I.T

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Abstract

This project is applying a Power Theft Detection System through Arduino. A potentiometer is employed in order to simulate power changes, and LEDs and a buzzer are used as indicators. The green LED will be on in a normal operating condition indicating safe usage. When the power consumption exceeds the established threshold (theft may occur), the green LED is turned off, and the red one is turned on, and a buzzer is used to create an alert. This cost effective and easy to implement system shows that microcontroller based monitoring can assist in the process of detecting unauthorized power consumption and minimizing power wastage.

Code

```
const int potentiometerPin = A0;
const int greenLEDpin = 10;
const int redLEDpin = 11;
const int buzzerPin = 12;

const int thresholdValue = 500;

void setup() {

    pinMode(greenLEDpin, OUTPUT);
    pinMode(redLEDpin, OUTPUT);
    pinMode(buzzerPin, OUTPUT);

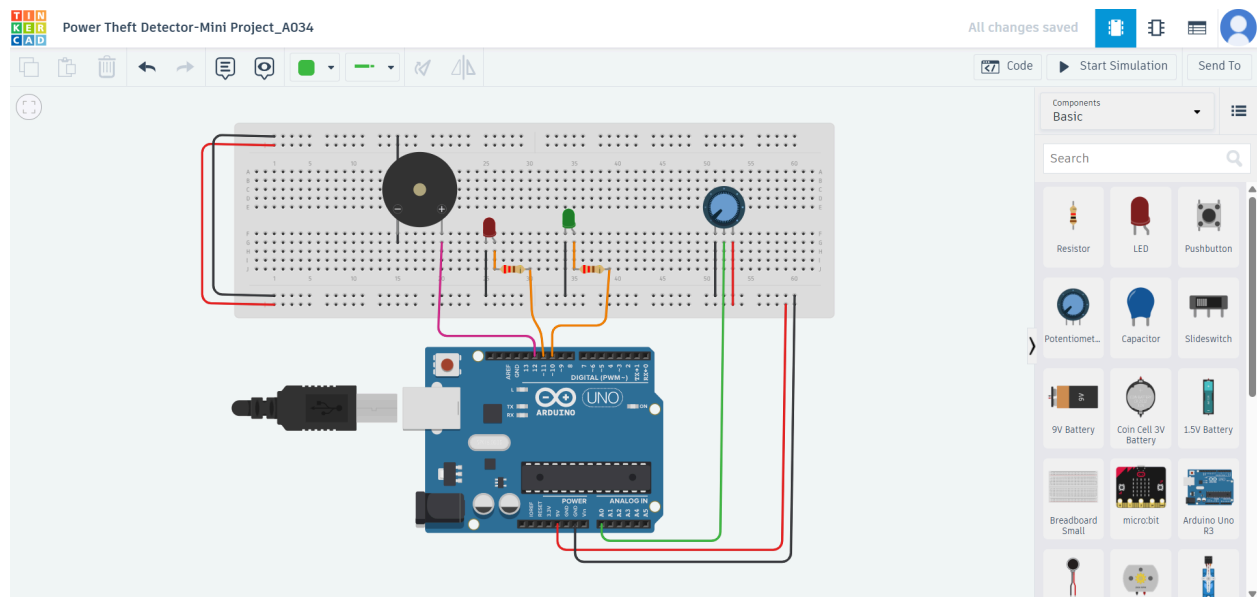
    digitalWrite(greenLEDpin, HIGH);
    digitalWrite(redLEDpin, LOW);
    digitalWrite(buzzerPin, LOW);
}

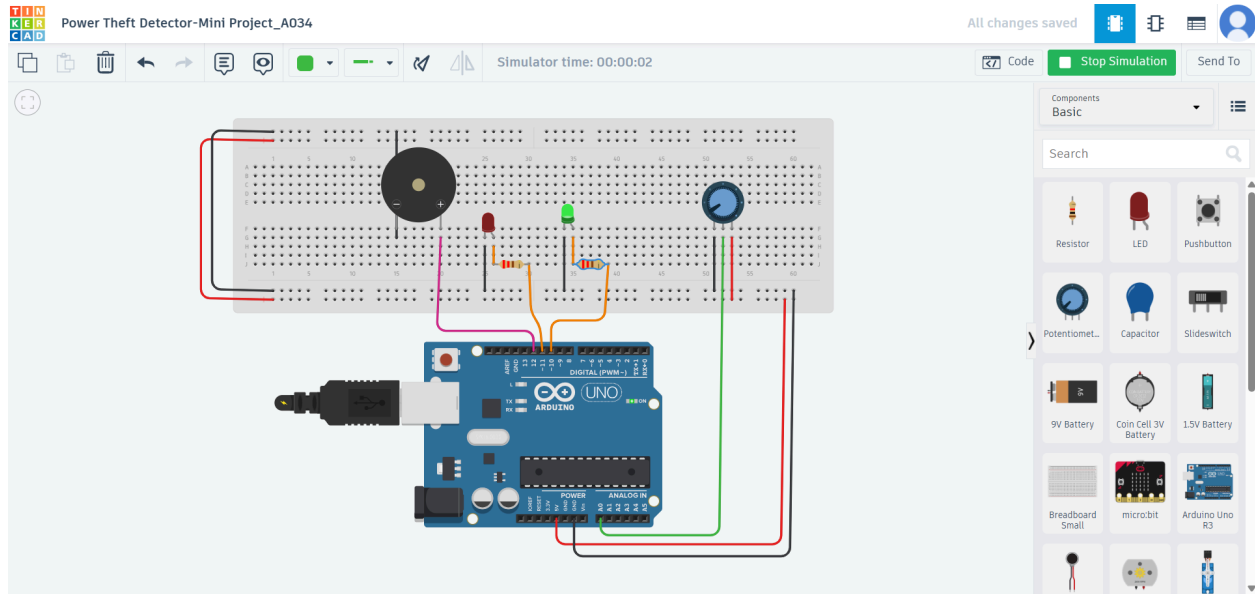
void loop() {
    int sensorValue = analogRead(potentiometerPin);

    if (sensorValue > thresholdValue) {
        digitalWrite(greenLEDpin, LOW);
        digitalWrite(redLEDpin, HIGH);
        tone(buzzerPin, 1000);
    } else {
        digitalWrite(greenLEDpin, HIGH);
        digitalWrite(redLEDpin, LOW);
        noTone(buzzerPin);
    }
}
```

```
delay(100);  
}
```

Diagram:



greenLED is ON - Normal Safe State***redLED is ON & Buzzer RINGS - Alerted Un-safe State***